

Microsoft Access V

Reports

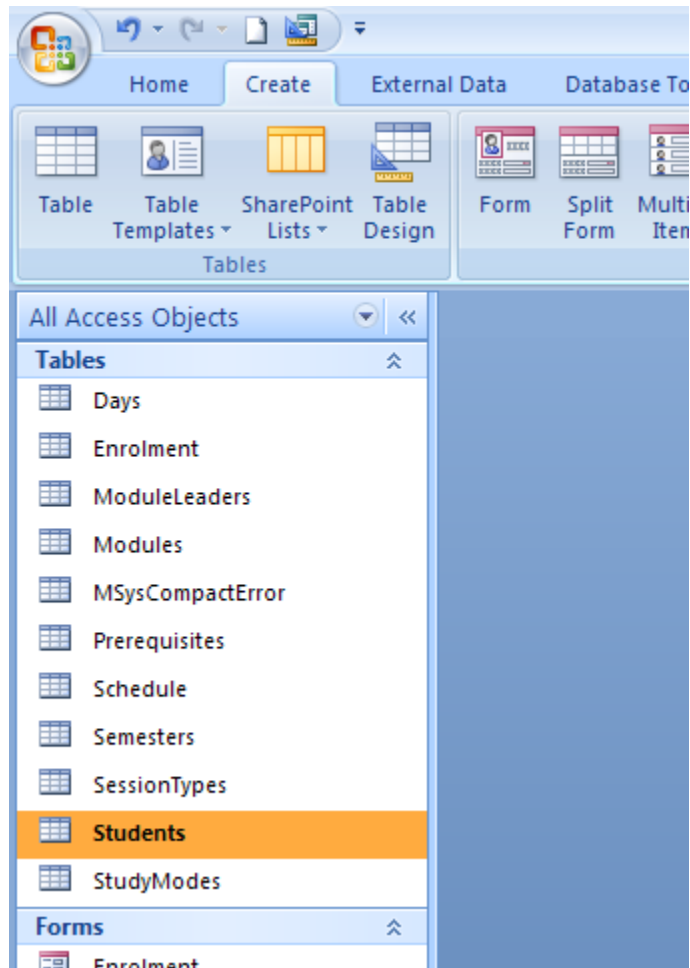
In the previous tutorial, we looked at creating an Access “Switchboard”, a menu-like form that makes it easy for your users to access *all* your forms. This helps turn an Access database into a useful application that can be employed by end-users. In this tutorial, we will look at creating reports.

Reports allow users to generate summaries or print-outs of data in the database. For many users, this is the whole reason to use database software – to generate meaningful reports that they can use to help run their business or organisation.

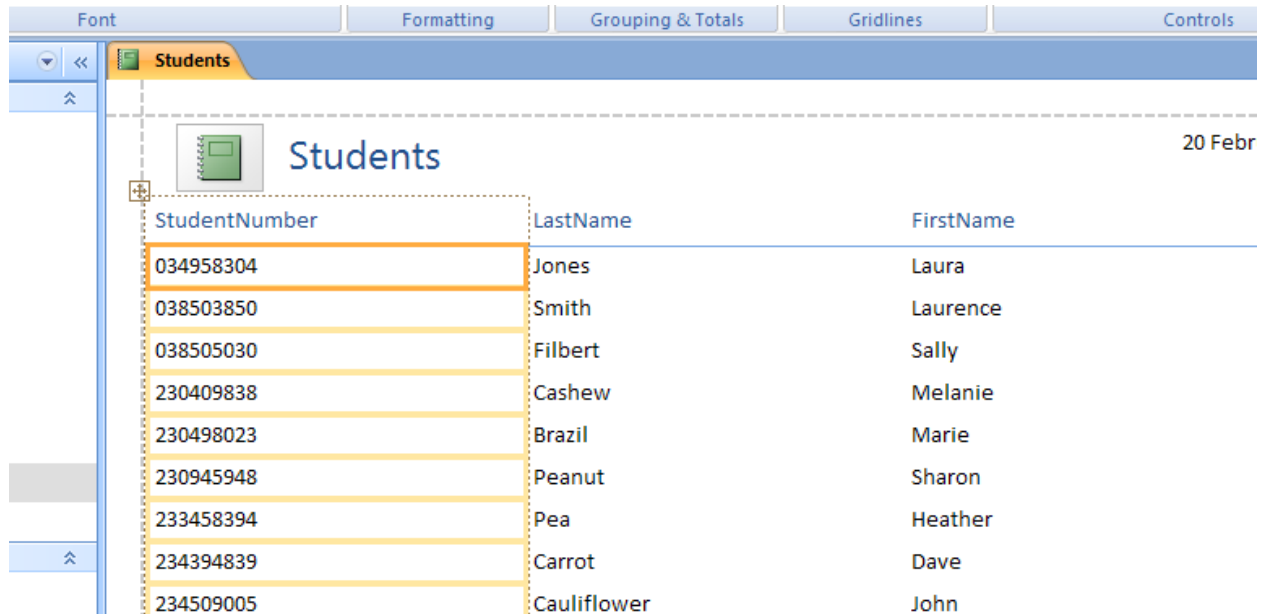
For this tutorial, you will use the same TutorialDatabase.mdb database that we used last week and in the previous tutorial. By now, it should have forms to manipulate the data in all the tables and a switchboard.

We’ll start by creating a very simple report.

Load the database, and highlight a table, such as Students.



Then click on the “Create” tab and click the “Report” icon in the “Reports” panel. A report will appear:



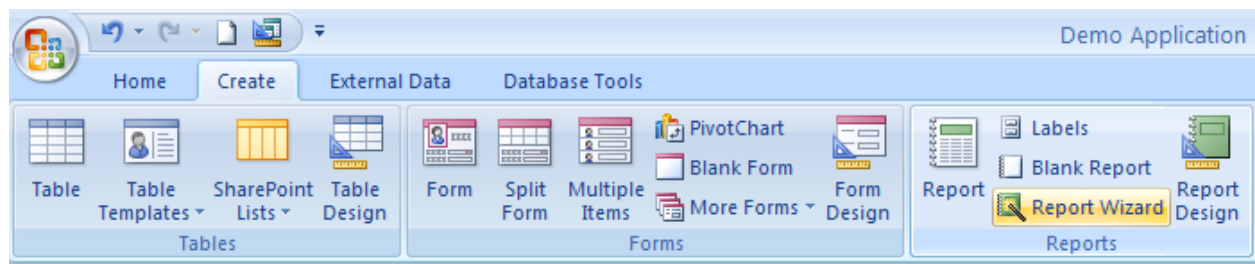
StudentNumber	LastName	FirstName
034958304	Jones	Laura
038503850	Smith	Laurence
038505030	Filbert	Sally
230409838	Cashew	Melanie
230498023	Brazil	Marie
230945948	Peanut	Sharon
233458394	Pea	Heather
234394839	Carrot	Dave
234509005	Cauliflower	John

Great! If you like, you can close the report and save its definition for later use.

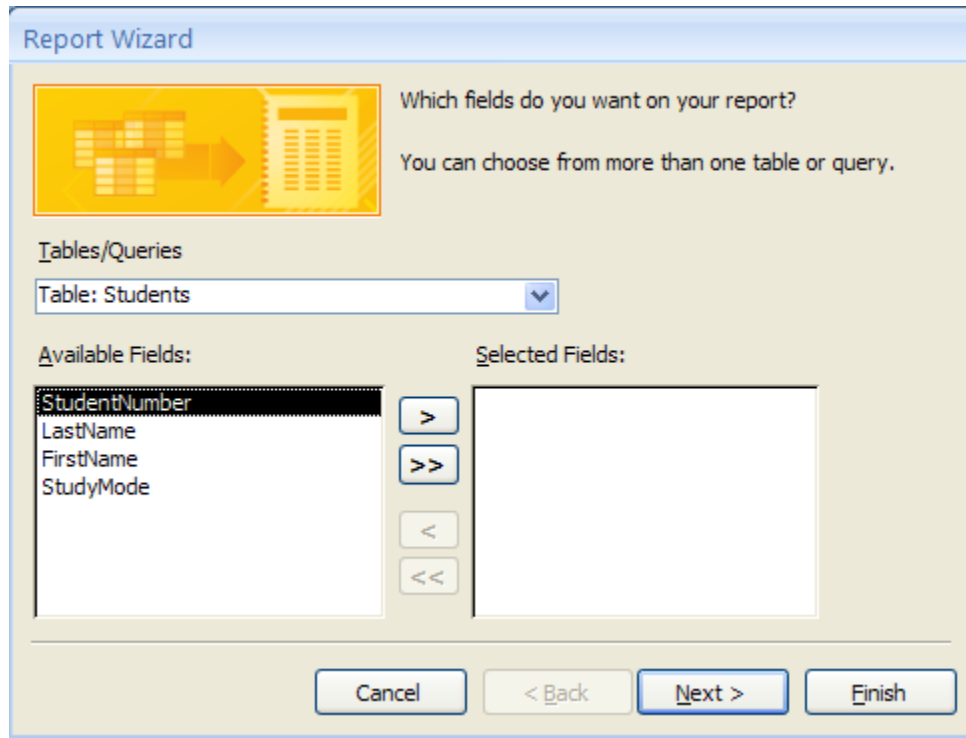
Unfortunately, such simple reports rarely meet user requirements. Usually, we have to at least modify the report’s layout, displayed columns, etc, before it’s useful.

Now we’ll try making a simple, but customisable report.

Highlight the Students table like before, but this time click on the “Report Wizard” icon, as shown below:



The “Report Wizard” dialogue box should appear:



Follow the instructions in the Wizard to create a simple report.

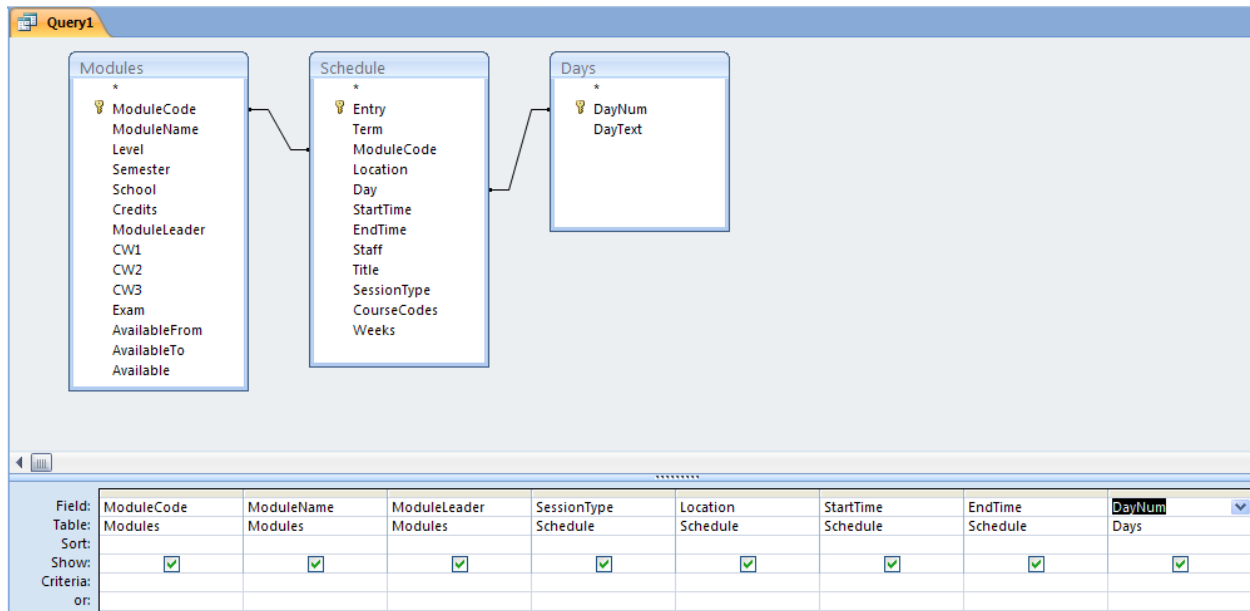
While this approach is, again, suitable for creating simple reports (which you can customise afterward in Design View), it has one serious limitation that we’ll see how to address in the remainder of this tutorial.

The serious limitation is that the report will present *all* the data in its associated table (or query – you can base reports on queries). Most practical reports, however, need to select a specified *range* of data. For example, we don’t want to display every invoice, but only those within a specified date range. Or, we don’t want to display every student, but only a specified student or group of students. Or, we don’t want to show the entire timetable, but only those modules running on a particular day.

In this example, we’re going to create a report that will show the modules running on a user-selected day of the week. While there are some simple, but user-unfriendly ways of doing this, we’re going to do it by creating a query, a form, and a report that will all work together to present a user-friendly way of presenting information.

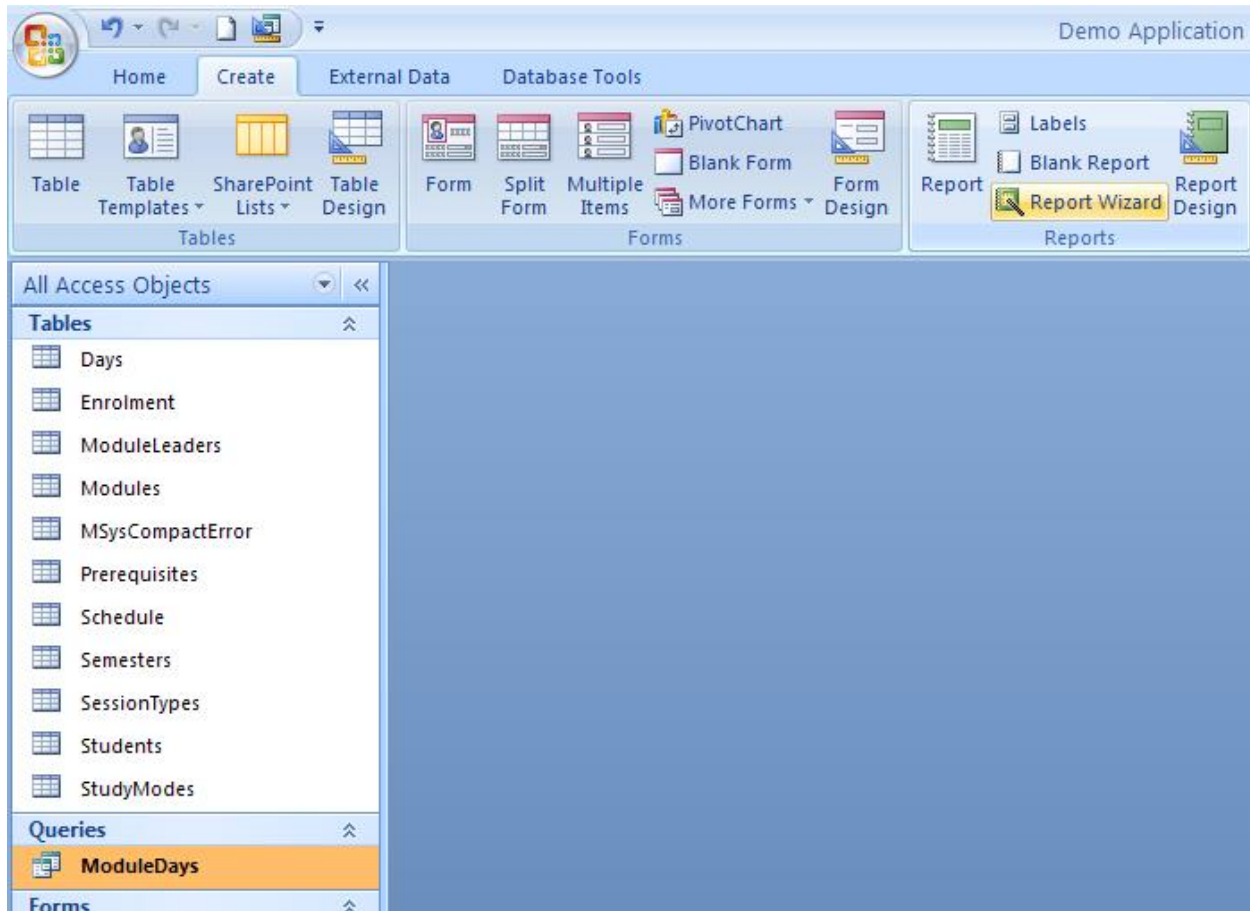
First, we need a query that will retrieve the data we wish to display.

Create the query shown below:

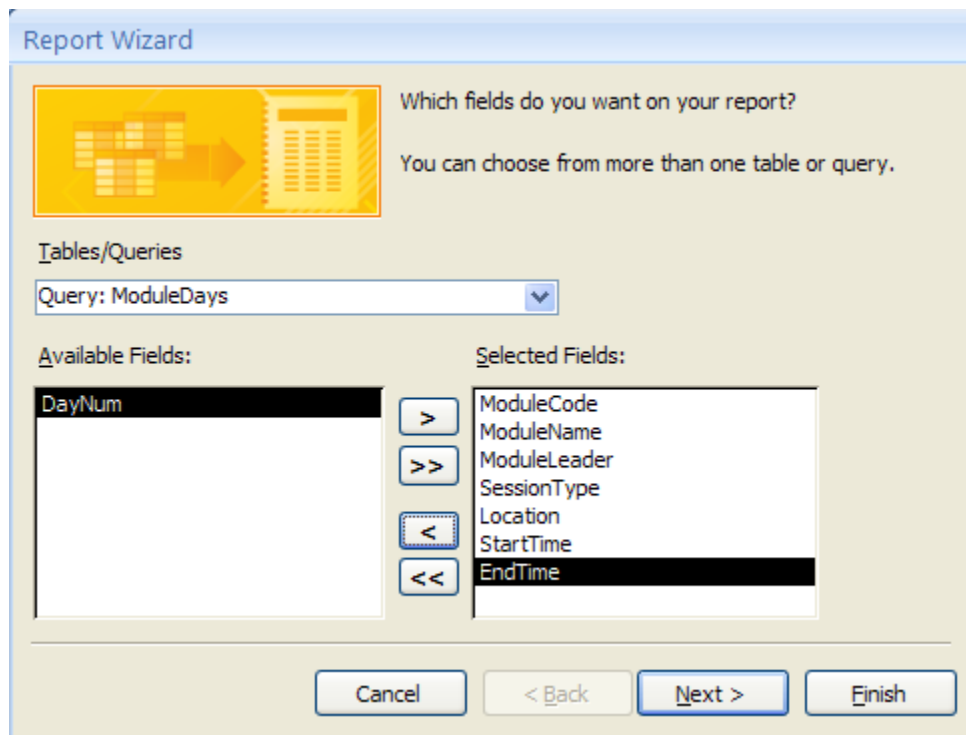


Save it with the name ModuleDays.

Now we'll create the report. Highlight the new ModuleDays query, and click the "Report Wizard" button in the "Reports" panel of the "Create" tab:



Use the Report Wizard to select all columns except DayNum. We don't need to display the day number on the report, because it would be the same throughout the report, as we're going to restrict it to a user-selected day.



The Report Wizard dialog box is titled "Report Wizard". It contains a yellow icon of a report and a question mark. The text "Which fields do you want on your report?" is displayed, followed by "You can choose from more than one table or query." Below this, the "Tables/Queries" section shows a dropdown menu with "Query: ModuleDays" selected. The "Available Fields:" list contains "DayNum". The "Selected Fields:" list contains "ModuleCode", "ModuleName", "ModuleLeader", "SessionType", "Location", "StartTime", and "EndTime". Navigation buttons include ">", ">>", "<<", "<", "Cancel", "< Back", "Next >", and "Finish".

Report Wizard

Which fields do you want on your report?

You can choose from more than one table or query.

Tables/Queries

Query: ModuleDays

Available Fields:

DayNum

Selected Fields:

ModuleCode
ModuleName
ModuleLeader
SessionType
Location
StartTime
EndTime

Cancel < Back Next > Finish

Press the “Next” button. Set one grouping level, on ModuleLeader:

The 'Report Wizard' dialog box is shown with the title 'Report Wizard'. The question 'Do you want to add any grouping levels?' is at the top. On the left, a list of fields includes 'ModuleCode', 'ModuleName', 'SessionType', 'Location', 'StartTime', and 'EndTime'. To the right of this list are four buttons: '>', '<', an up arrow, and a down arrow, with the label 'Priority' below them. On the right side, a box titled 'ModuleLeader' contains the text 'ModuleCode, ModuleName, SessionType, Location, StartTime, EndTime'. At the bottom, there are five buttons: 'Grouping Options ...', 'Cancel', '< Back', 'Next >', and 'Finish'.

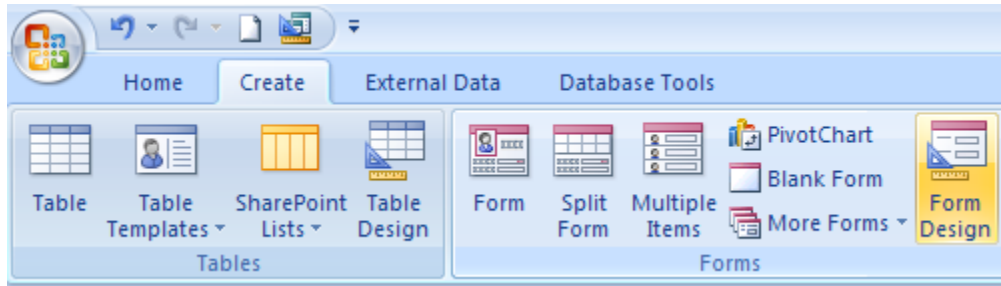
Press the “Next” button. Sort the report by ModuleCode:

The 'Report Wizard' dialog box is shown with the title 'Report Wizard'. The question 'What sort order and summary information do you want for detail records?' is at the top. On the left, there is a yellow graphic of a spiral notebook with a table and a double-headed vertical arrow. To the right, a text box says 'You can sort records by up to four fields, in either ascending or descending order.' Below this, there are four rows of input fields. Row 1 has 'ModuleCode' in the first dropdown and 'Ascending' in the second. Rows 2, 3, and 4 have empty dropdowns and 'Ascending' in the second column. Below these is a 'Summary Options ...' button. At the bottom, there are four buttons: 'Cancel', '< Back', 'Next >', and 'Finish'.

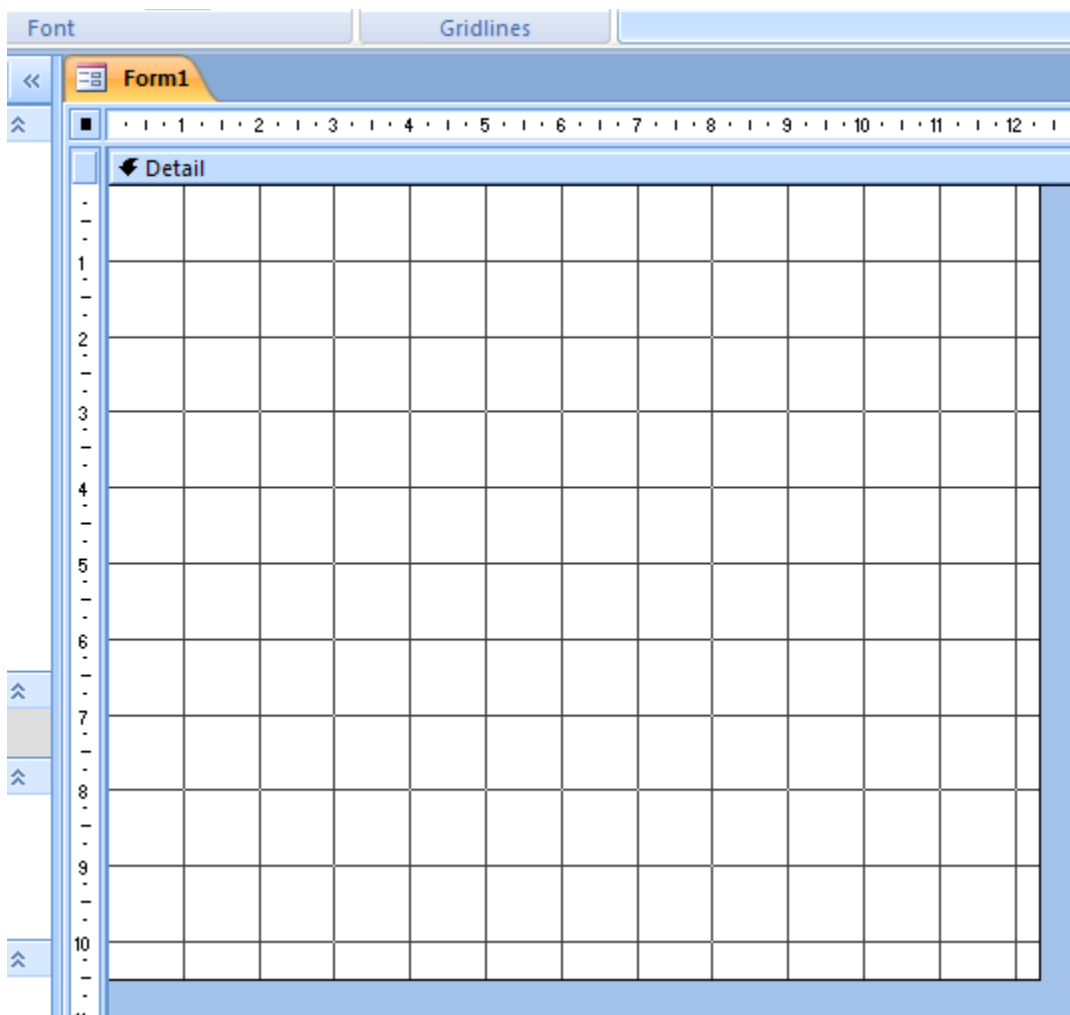
Press the “Finish” button. Save the report under the name “ModuleDays”.

Now we’ll create a form that will be used to pick the day we wish to display, and launch the report.

Click on the “Blank form” button in the “Forms” panel of the “Create” tab:

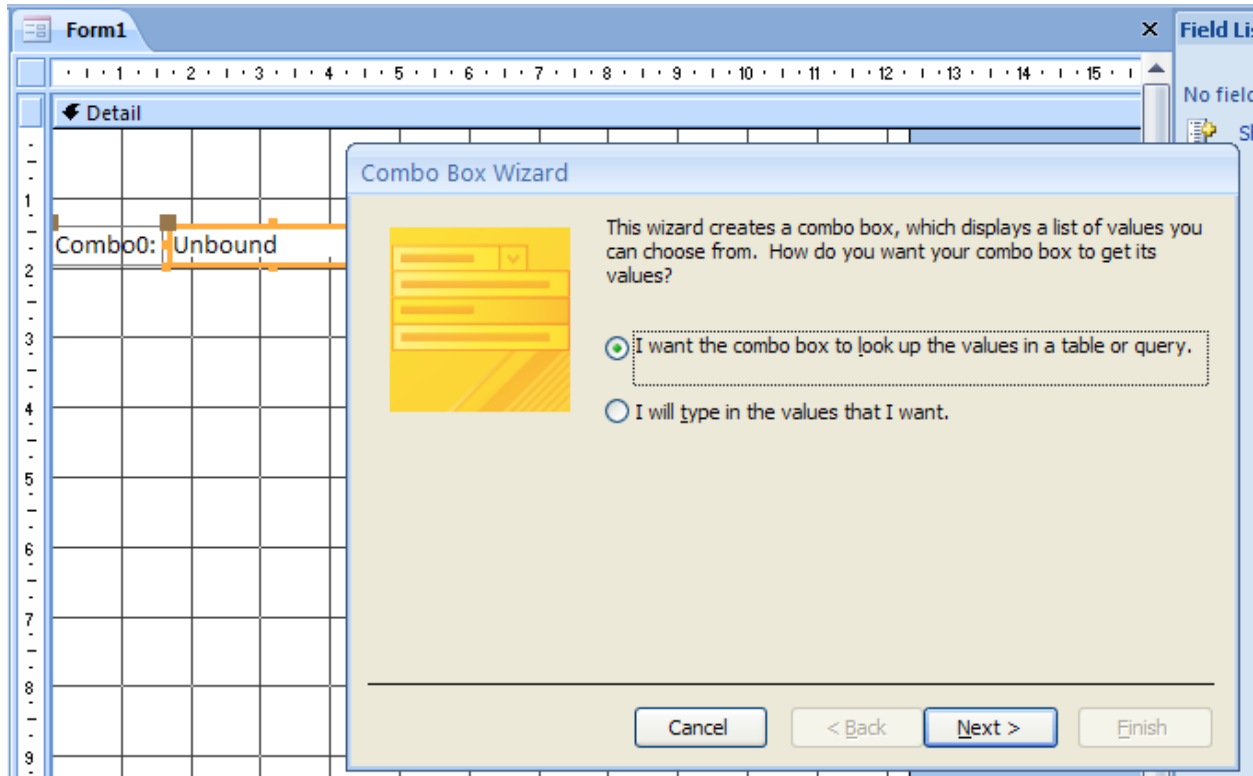


You should see a blank form canvas like the following:

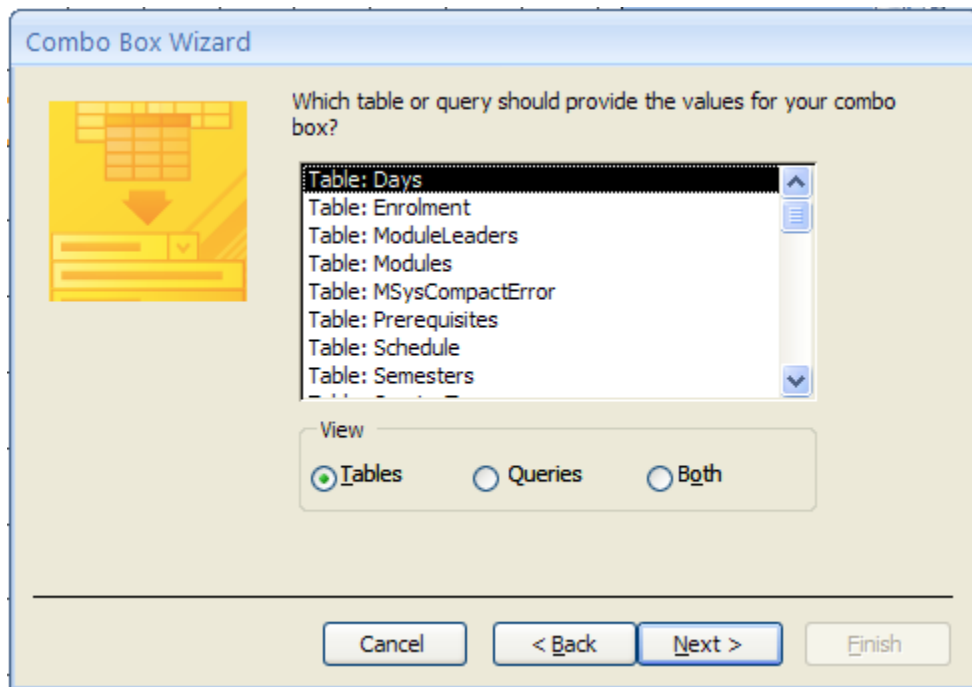


First, we need a combobox that will let us pick the day we wish to display on the report.

Select a combobox from the “Controls” panel, then click on the form canvas in roughly the place where you wish to put it. The “Combo Box Wizard” should appear:



Pick the first radiobutton option, as shown above, and press the “Next” button. Pick the “Days” table, as shown below:



Then press “Next”:



Select the DayNum and DayText columns as shown above, and press the “Next” button:

The Combo Box Wizard dialog box is titled "Combo Box Wizard". It contains the text: "What sort order do you want for the items in your list box?" and "You can sort records by up to four fields, in either ascending or descending order." Below this, there are four rows of configuration. Row 1 has a dropdown menu with "DayText" selected and an "Ascending" button. Rows 2, 3, and 4 have empty dropdown menus and "Ascending" buttons. At the bottom, there are four buttons: "Cancel", "< Back", "Next >", and "Finish".

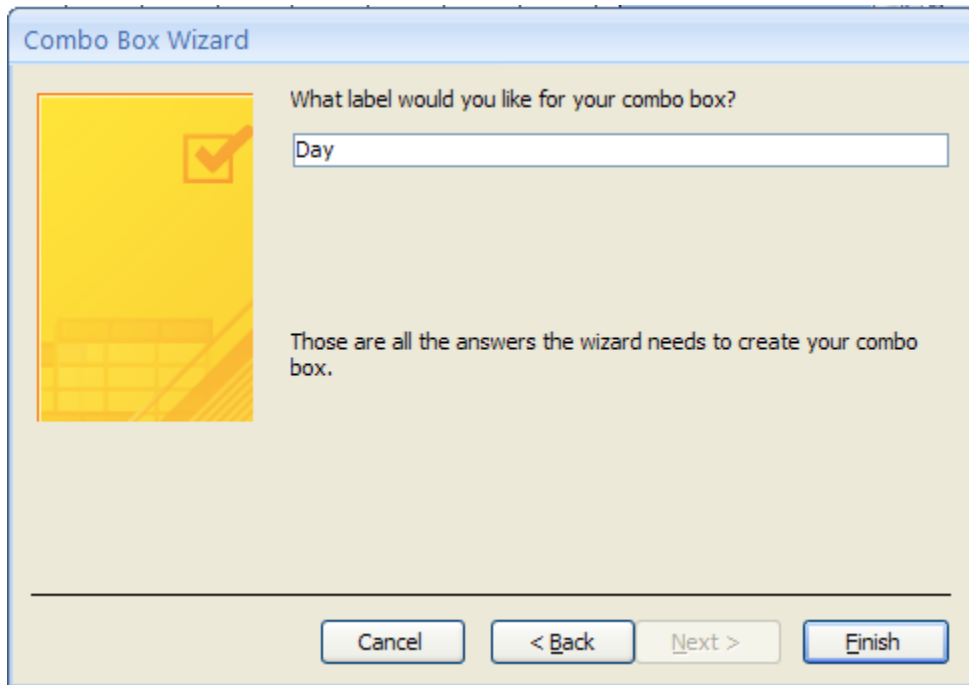
Field	Sort Order
1 DayText	Ascending
2	Ascending
3	Ascending
4	Ascending

Sort by DayText, as shown above. Comboboxes should always present sorted values. Press the “Next” button:

The Combo Box Wizard dialog box is titled "Combo Box Wizard". It contains the text: "How wide would you like the columns in your combo box?" and "To adjust the width of a column, drag its right edge to the width you want, or double-click the right edge of the column heading to get the best fit." Below this, there is a checkbox labeled "Hide key column (recommended)" which is checked. Below the checkbox is a table with a header row "DayText" and seven data rows: "Fri", "Mon", "Sat", "Sun", "Thu", "Tue", and "Wed". At the bottom, there are four buttons: "Cancel", "< Back", "Next >", and "Finish".

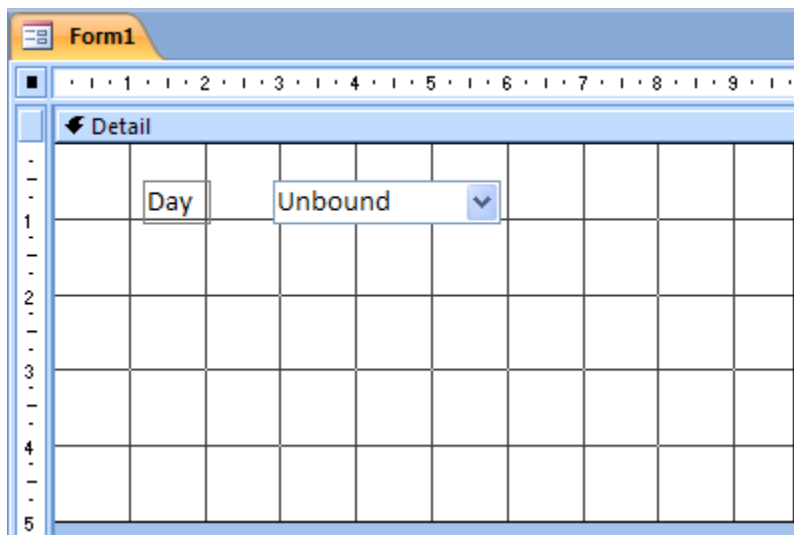
DayText				
Fri				
Mon				
Sat				
Sun				
Thu				
Tue				
Wed				

Be sure the “Hide key column” option is selected. This will hide the day number (not particularly friendly for your users), and only display the text. Press the “Next” button:



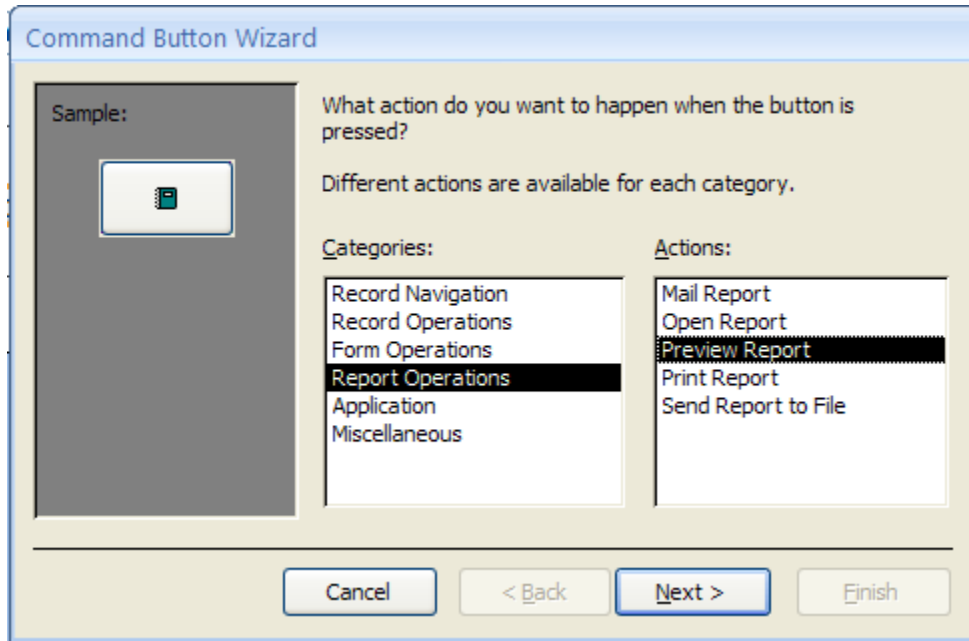
The image shows a 'Combo Box Wizard' dialog box. On the left is a yellow square with a red checkmark and a faint grid pattern. The main text area contains the question 'What label would you like for your combo box?' followed by a text box containing the word 'Day'. Below this, it says 'Those are all the answers the wizard needs to create your combo box.' At the bottom are four buttons: 'Cancel', '< Back', 'Next >', and 'Finish'.

Enter the text that will appear as a prompt on the combo box, as shown above. Then press the “Finish” button. So far, your form should look like the following:



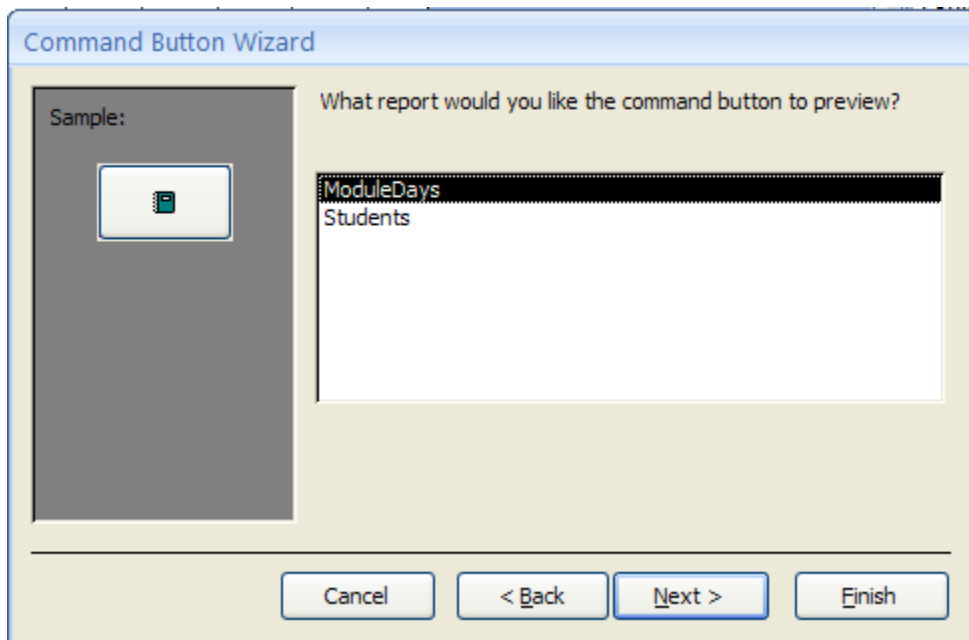
The image shows a screenshot of a form titled 'Form1'. It features a grid with 5 rows and 9 columns. The first row is highlighted. In the first column of the first row, the text 'Day' is entered. In the second column of the first row, a dropdown menu is open, showing the text 'Unbound' and a downward arrow. The grid is labeled 'Detail' at the top left.

Now we'll need a button that will launch the report. Put a button on the form, and the "Command Button Wizard" should appear:

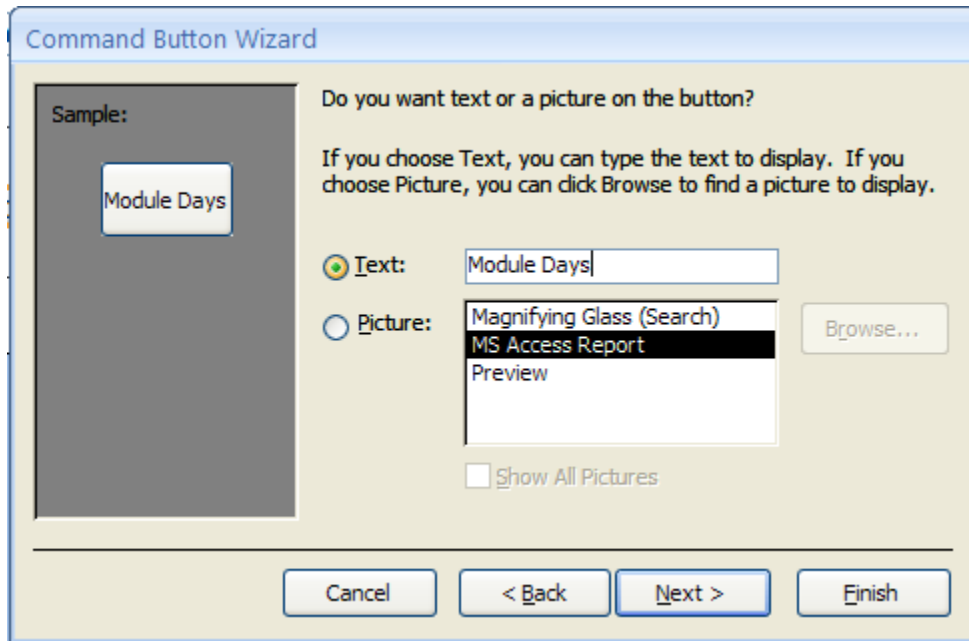


Select "Report Operations" under "Categories", and "Preview Report" under "Actions".

Then click the "Next" button:



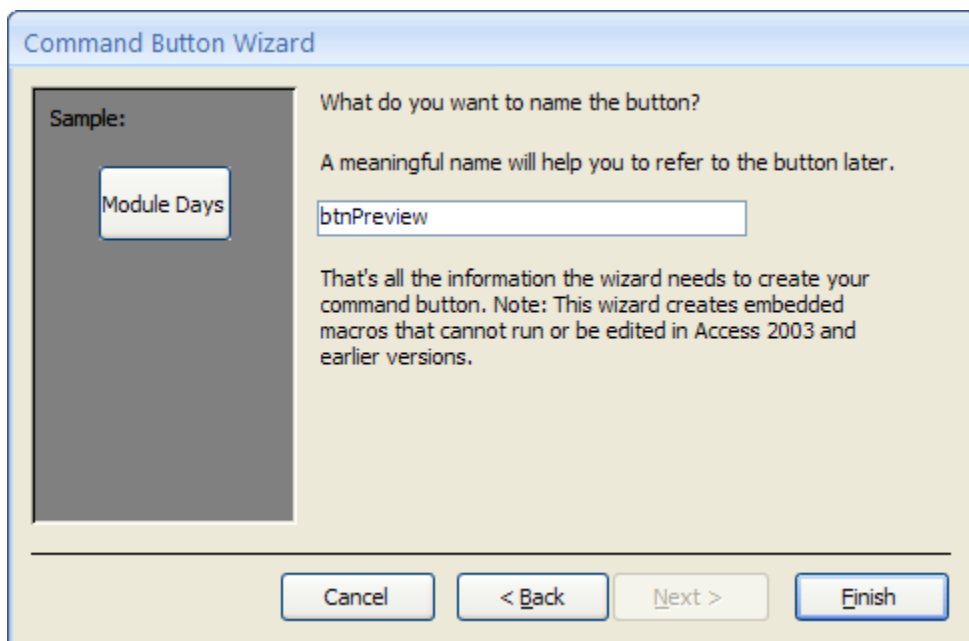
Select the “ModuleDays” report as shown above, then click the “Next” button:



The screenshot shows the "Command Button Wizard" dialog box. On the left, a "Sample:" area displays a button labeled "Module Days". The main text asks, "Do you want text or a picture on the button?". Below this, it explains: "If you choose Text, you can type the text to display. If you choose Picture, you can click Browse to find a picture to display." The "Text" radio button is selected. The text input field contains "Module Days". The "Picture" radio button is unselected, and its dropdown menu is open, showing "Magnifying Glass (Search)", "MS Access Report", and "Preview". A "Browse..." button is next to the dropdown. At the bottom, there is a "Show All Pictures" checkbox and navigation buttons: "Cancel", "< Back", "Next >", and "Finish".

Choose “Text”, and enter “Preview Report”, as shown above. This is the text that will appear on the button.

Then press the “Next” button.



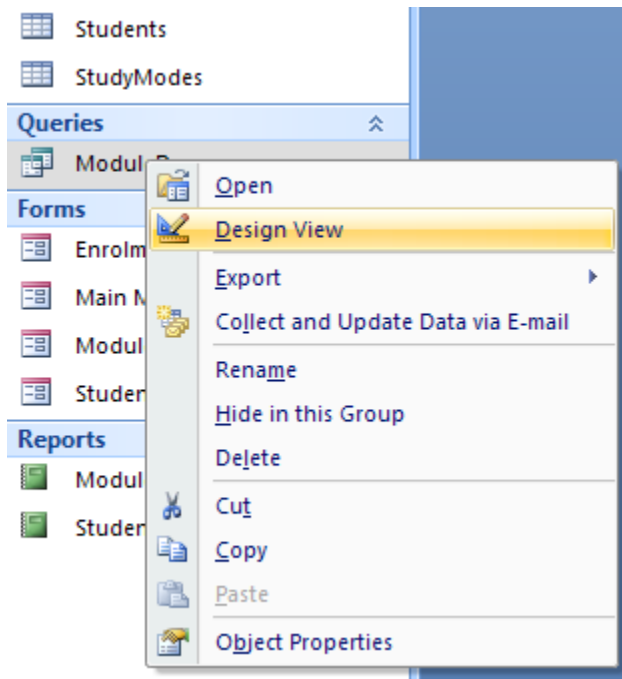
The screenshot shows the "Command Button Wizard" dialog box at the next step. The "Sample:" area still shows the "Module Days" button. The main text asks, "What do you want to name the button?". Below this, it says: "A meaningful name will help you to refer to the button later." The input field contains "btnPreview". A note at the bottom states: "That's all the information the wizard needs to create your command button. Note: This wizard creates embedded macros that cannot run or be edited in Access 2003 and earlier versions." Navigation buttons at the bottom are "Cancel", "< Back", "Next >", and "Finish".

Give the button a meaningful name, as shown above. Then click “Finish”.

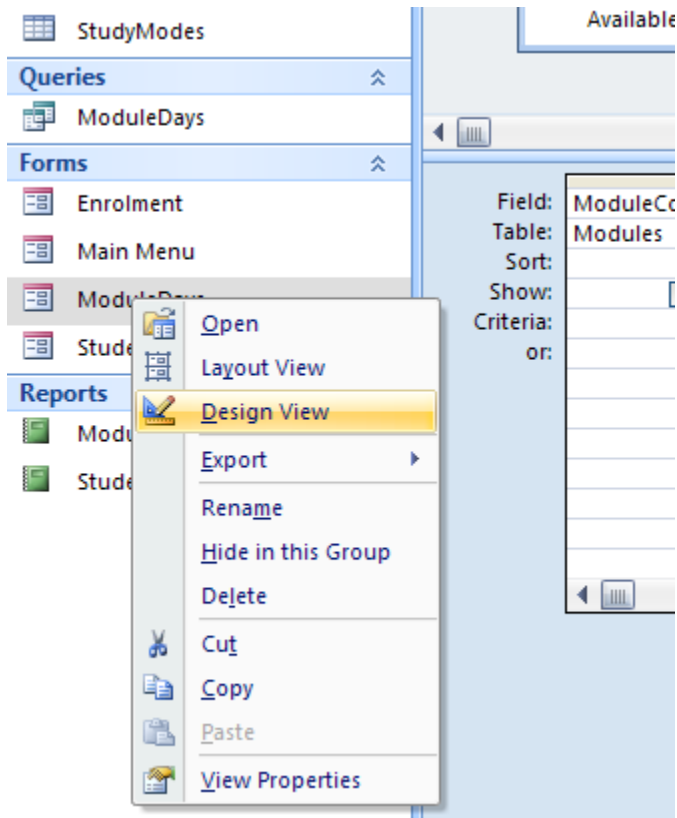
Close the form, and save it under the name “ModuleDays”.

Now we'll do the final steps to link the query to the combobox on the form you created, so that the query will only retrieve the rows that match the day selected via the combobox.

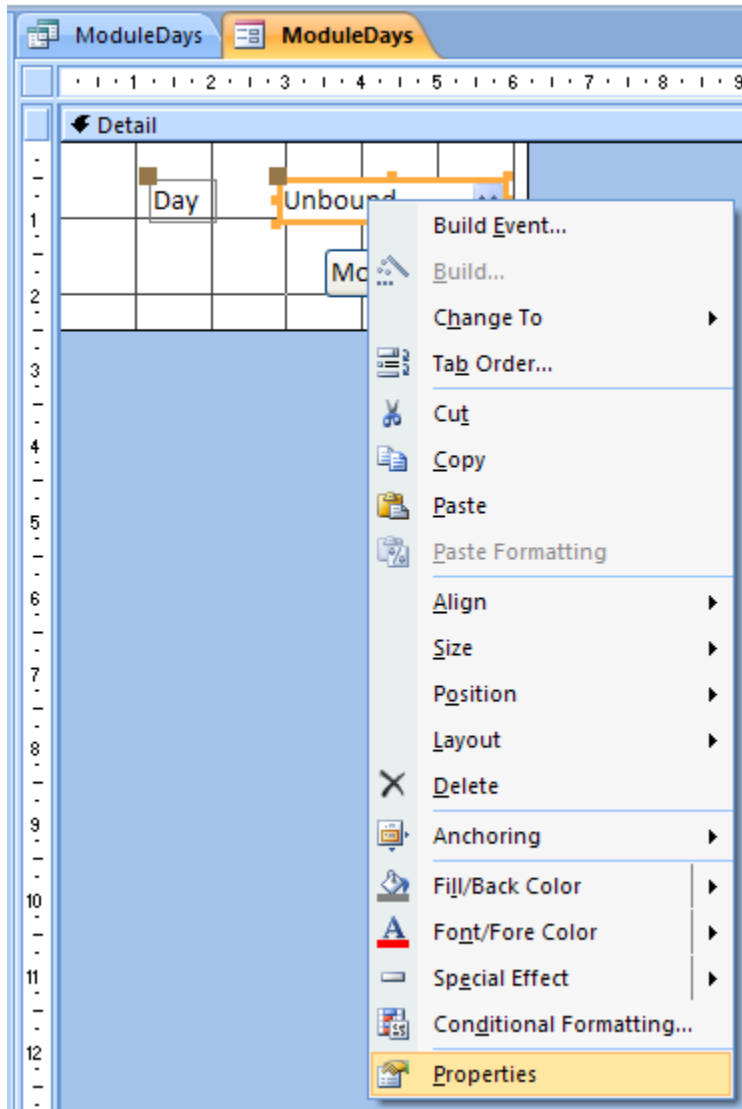
Right-click the “ModuleDays” query, and select “Design View”, as shown below:



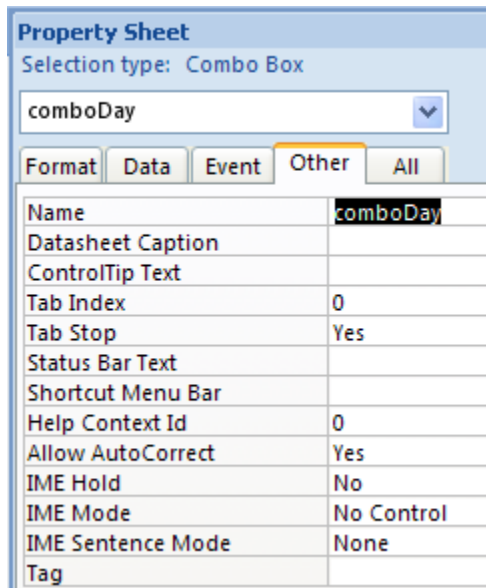
Leave the query's design view open in its tab. Right click on the "ModuleDays" form and select "Design View", as shown below:



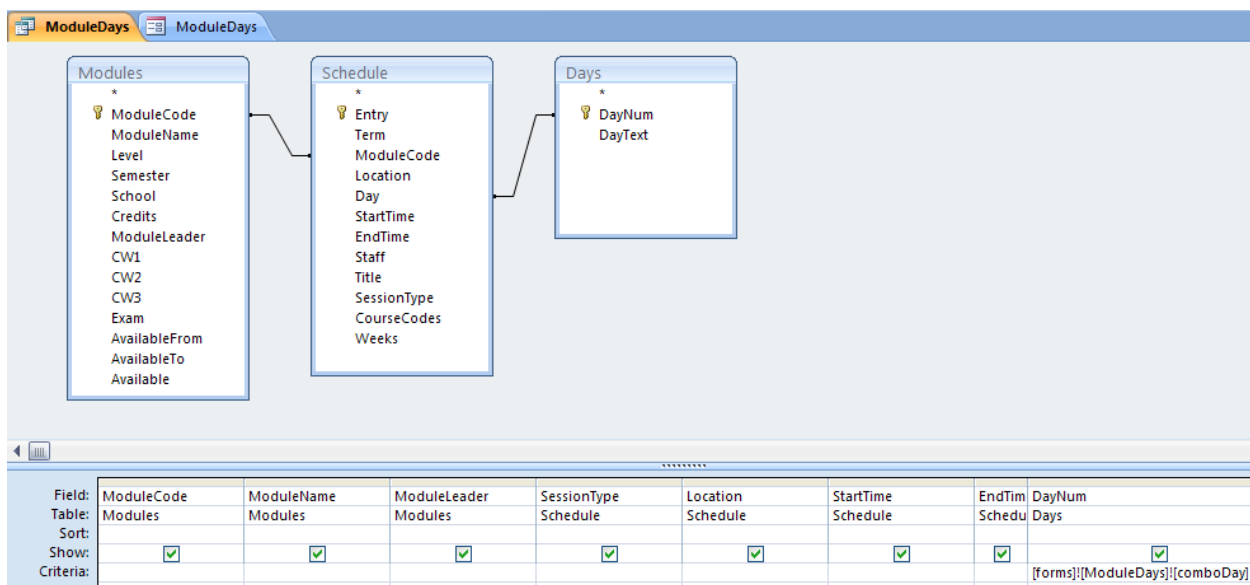
Now we need to give the combobox a name that can be referenced in the query. Right-click on the combobox and select Properties, as shown below:



Use the Property Sheet to set the combobox's name to "comboDay":



Now we'll alter the "ModuleDays" query so that it is restricted to rows that match the current value in the "comboDay" combobox in the "ModuleDays" form. Switch to the "ModuleDays" query, as shown below:



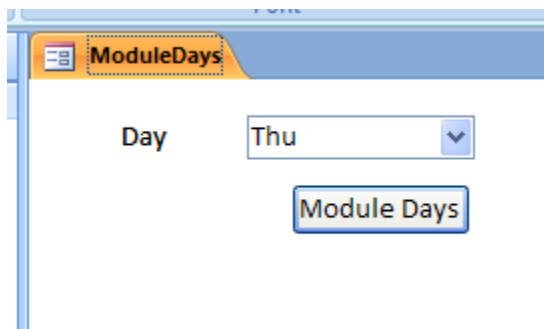
In the "Criteria" cell of the "DayNum" column, put the following:

[Forms]![ModuleDays]![comboDay]

This tells the query to only retrieve rows that match the current value of the "comboDay" combobox in the "ModuleDays" form.

Now close and save the “ModuleDays” query and the “ModuleDays” form.

Now let’s try the report! Double-click on the “ModuleDays” form to open it:



Select a day, as shown above, then press the button to display the report:

ModuleDays					
ModuleLeader	ModuleCode	ModuleName	SessionType	Location	artTime
Jones (Clifton)					
	5CC073	Operating System	Tutorial	{KR - B204}	10
	5CC073	Operating System	Lecture	{KR - E603}	9
	5CC073	Operating System	Tut/Practical	{KR - B204}	16
	5CC073	Software Develop	Lecture	{KR - E604}	13
	5CC073	Software Develop	Tut/Practical	{KR - B204}	16
Parkes (Dennis)					
	5CC048	User Centred Syst	Tut/Practical	{KR - N109}	13
Richard Self					
	6CC044	IT Governance an	Lecture	{KR - N512}	10
	6CC044	IT Governance an	Tut/Practical	{KR - E603}	11
	6CC044	IT Governance an	Tut/Practical	{KR - N103}	12
Ripplin (Wayne)					
	4CC044	Computer Archite	Tut/Practical	{KR - B202}	16
	4CC044	Computer Archite	Tut/Practical	{KR - B202}	17
	4CC050	Software Develop	Tut/Practical	{KR - B202}	14
	4CC050	Software Develop	Tut/Practical	{KR - B204}	14

You can use the “Design View” of the report to alter its layout, title, column headings, and so on.

Finally, you should alter your Switchboard to provide a button that will invoke the “ModuleDays” report’s form:

The screenshot shows a Microsoft Access form titled "Main Menu". The form has a tabbed interface with three tabs: "ModuleDays", "ModuleDays", and "Main Menu". The "Main Menu" tab is currently selected. The form is divided into three sections: "Form Header", "Detail", and "Form Footer". The "Form Header" section contains a text box labeled "Main Menu". The "Detail" section is a grid with 12 columns and 3 rows. The first row of the grid contains a text box labeled "Enrolment". The "Form Footer" section is empty.

Main Menu											

Task

Create a report called “Class Lists” that generates class lists. It should show the following:

- Student number, last name, first name,

- Module code, module name, location, start time, end time, and

- Module leader’s name, email address, and phone number.

The user should be able to specify the day, module level, and term, for which the Class Lists will be generated. You should define the report grouping levels so that the module code and module leader’s information is not repeated for every student.